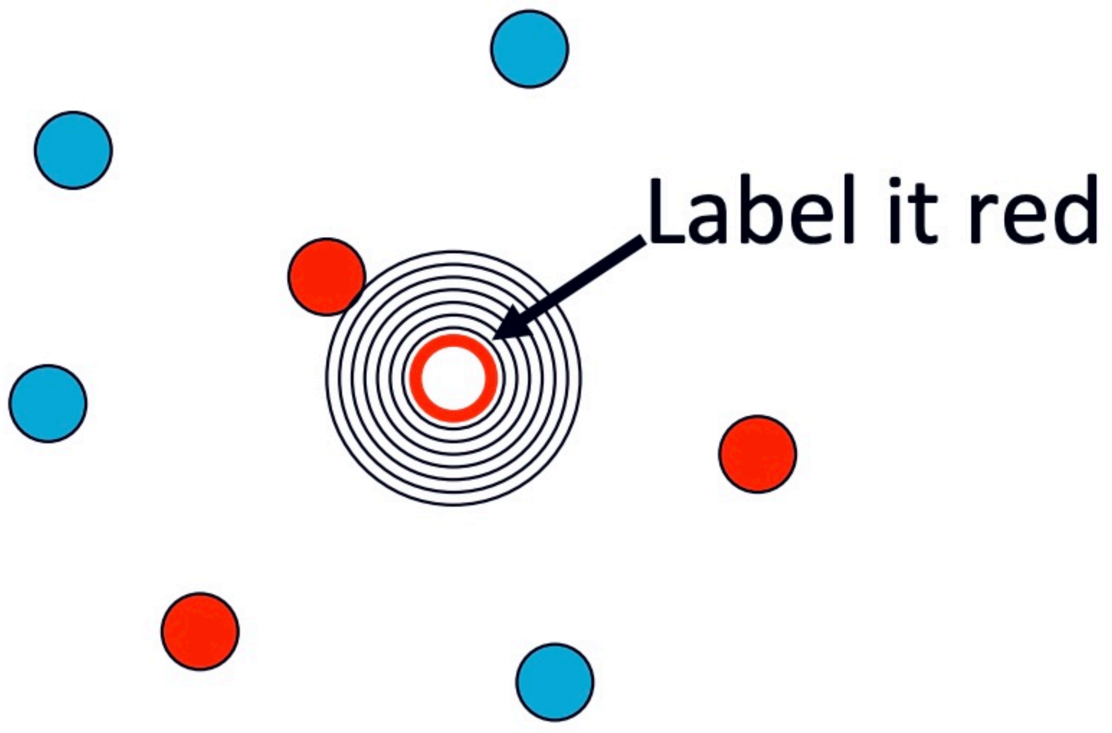
Classification: Part II

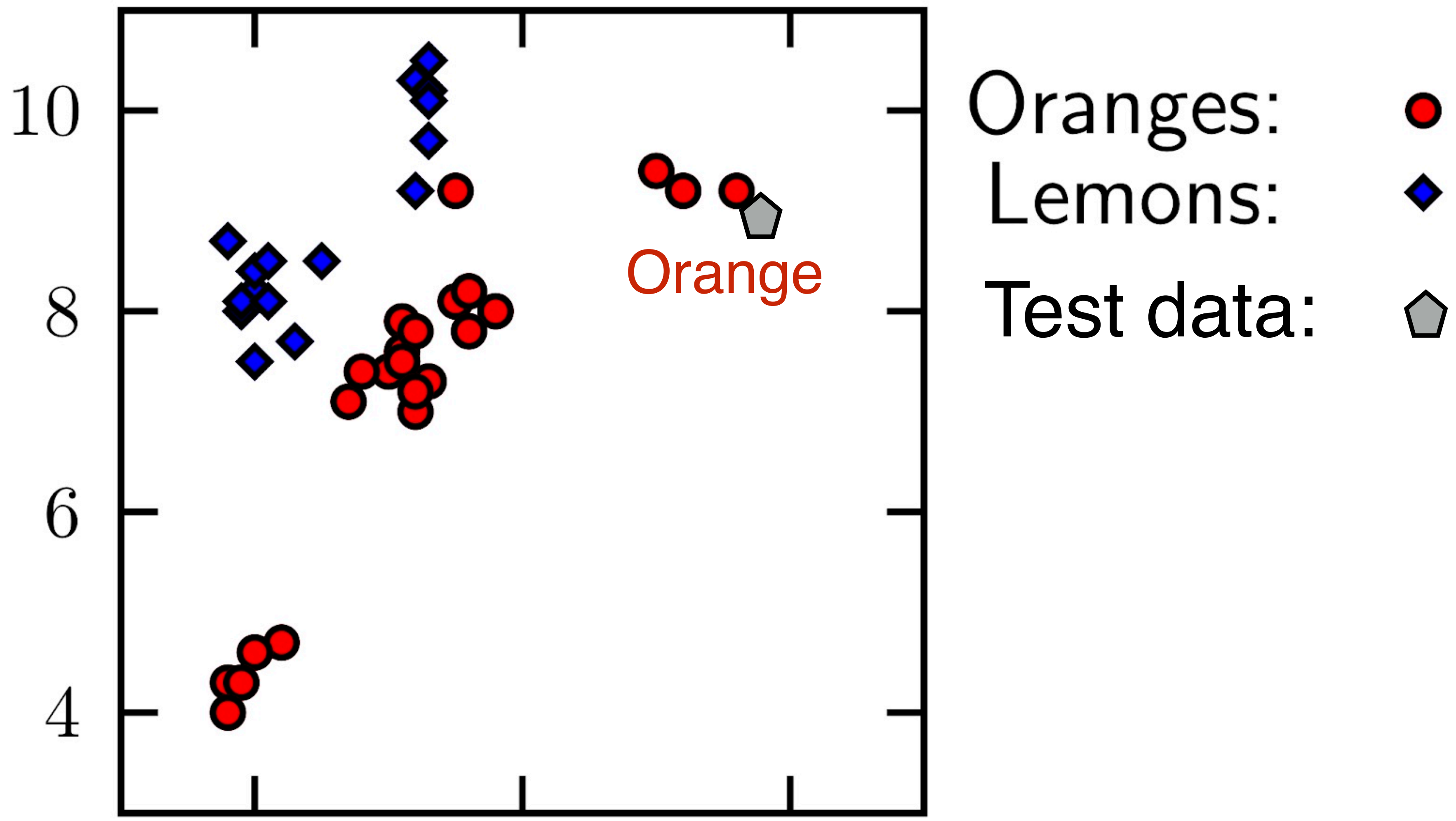
Some slides are adapted from those of Dr. Ke Yuan

Nearest Neighbour

- Simplest of all classifiers: **1-Nearest Neighbour**
- **Simple idea:** Label a new sample the same as its closest data point



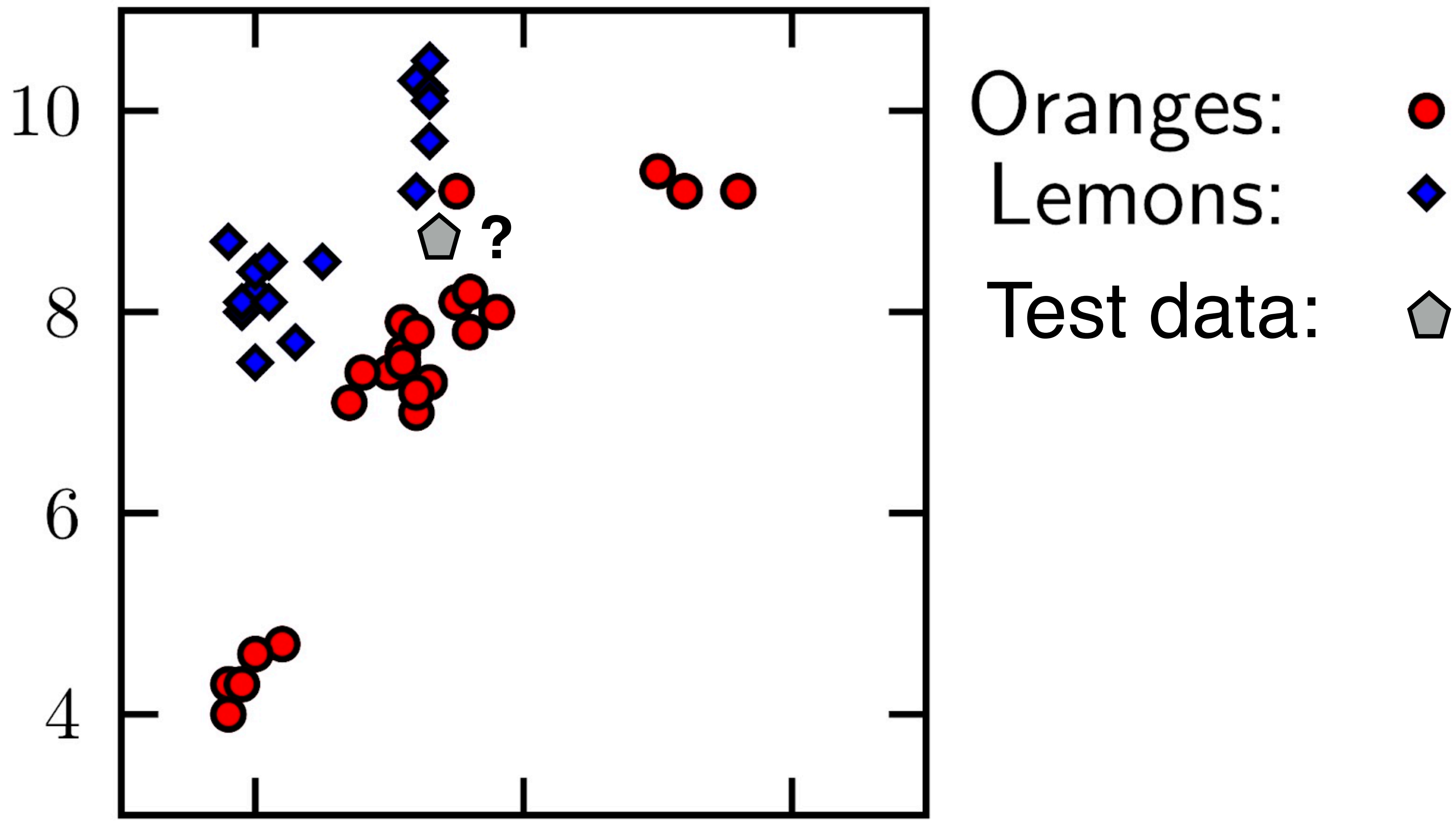
1-Nearest Neighbour



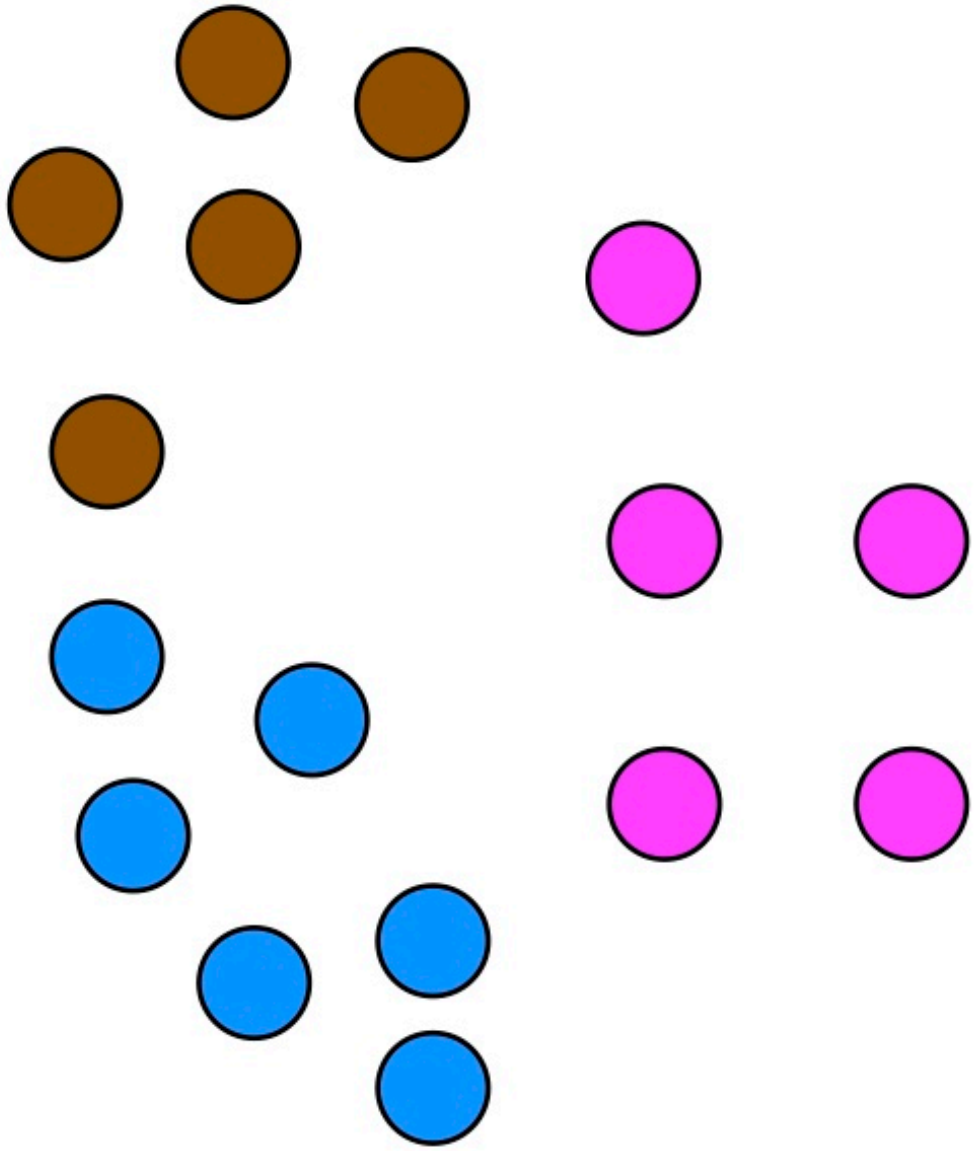
$$y_{new} \leftarrow y_{i*}$$

$$i* = \underset{i}{\operatorname{argmin}}\ dist(\mathbf{x}_i, \mathbf{x}_{new})$$

1-Nearest Neighbour

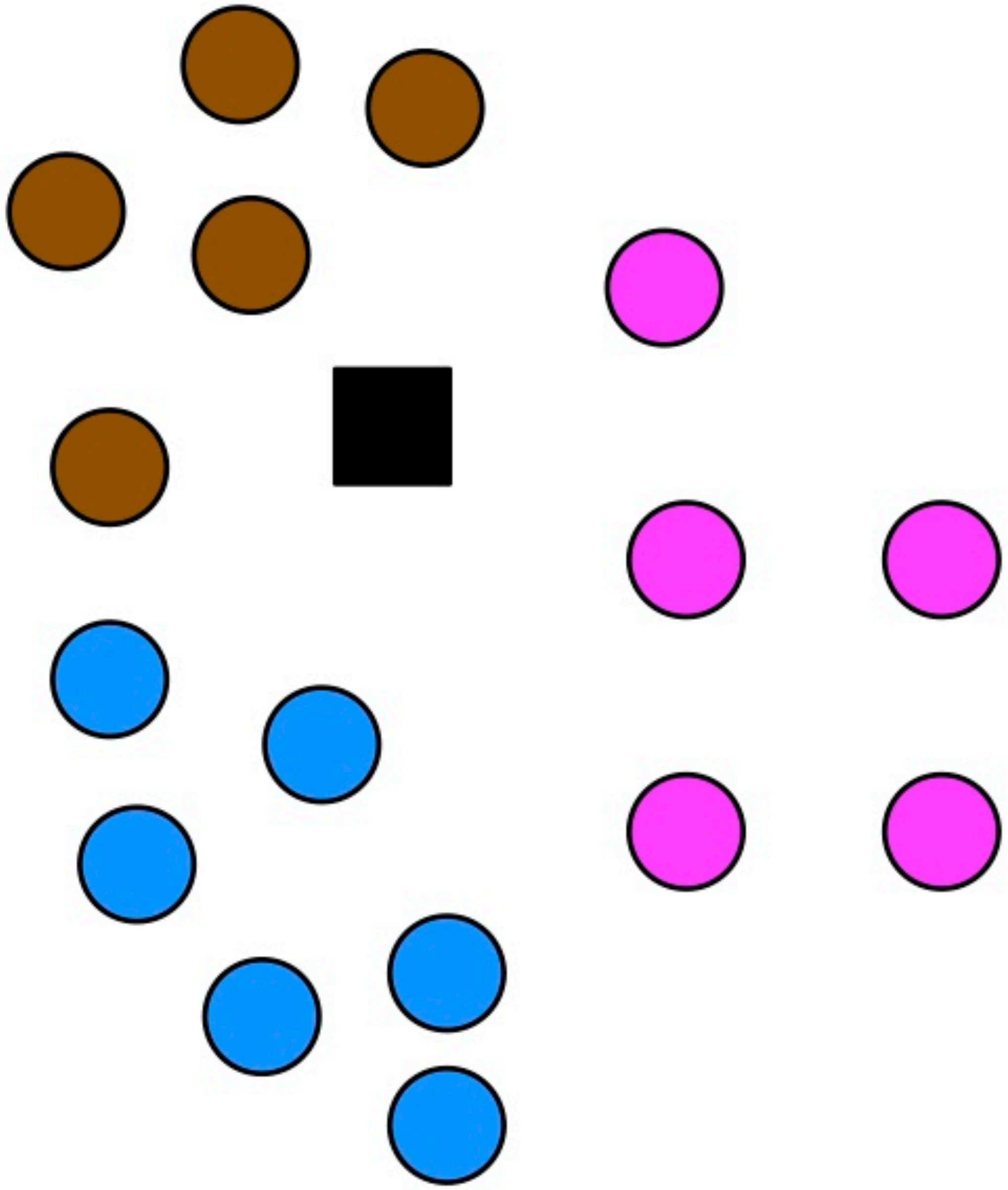


K-Nearest Neighbour



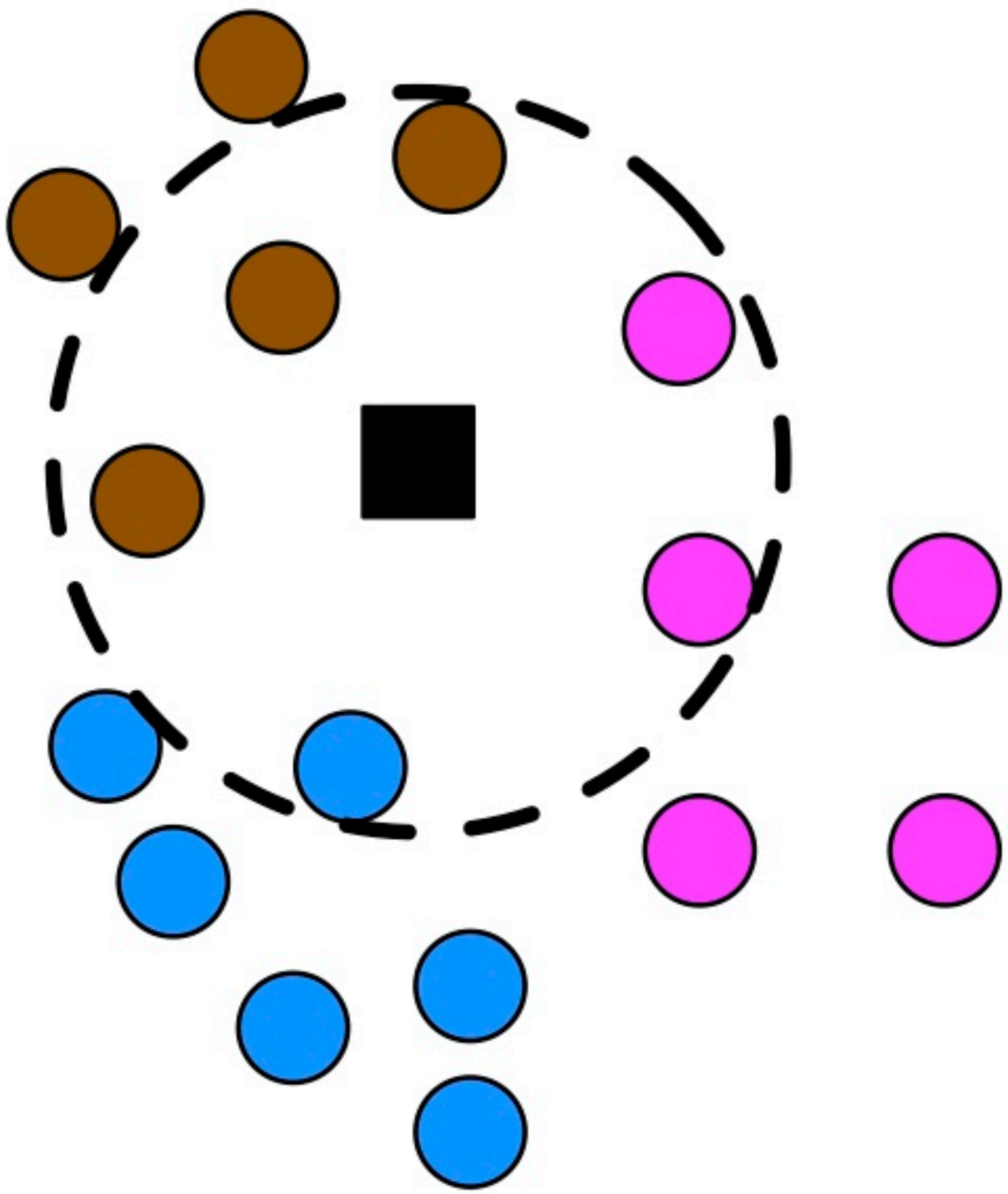
Training data from 3 classes.

K-Nearest Neighbour



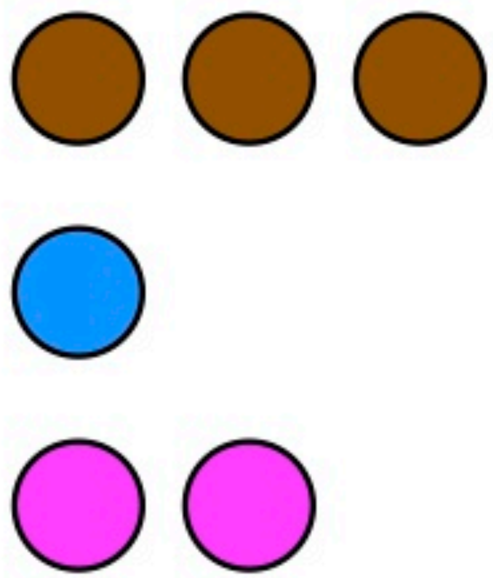
Test point.

K-Nearest Neighbour



Find $K = 6$ nearest neighbours.

K-Nearest Neighbour



3 from class 1

1 from class 2

2 from class 3

Class one has most votes – classify $\mathbf{x}_{new}$ as belonging to class 1.

Support Vector Machines

Ali Gooya

Introduction

- We assume linearly separable data for binary classification.
- Training data $\mathrm{x}_1, \cdots, \mathrm{x}_N$, with target variables $\mathrm{t}_1, \cdots, \mathrm{t}_N$, $\mathrm{t}_i \in \{-1,1\}$
- We view the sign of for classification: $h(\mathrm{x}) = \mathrm{sign}(y(\mathrm{x}))$
- Note that we can scale w and b without changing $h(\mathrm{x})$
- Choose w and b so that for the closest point to the decision boundary we satisfy: $y(\mathrm{x}) = \pm 1$
- Therefore we assume:
    - For any nearest point x_1 with $t_1 = -1$, we have $\mathrm{w}^T\mathrm{x}_1 + b = -1$
    - For any nearest point x_2 with $t_2 = 1$, we have $\mathrm{w}^T\mathrm{x}_2 + b = -1$

Introduction

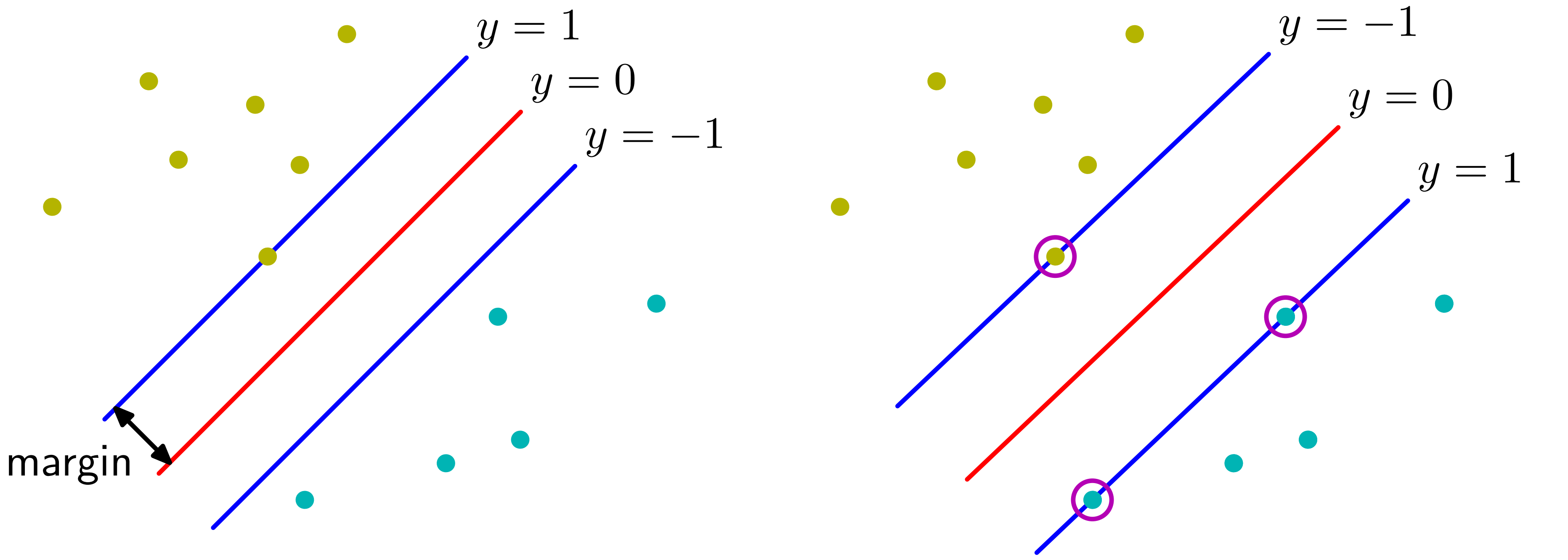
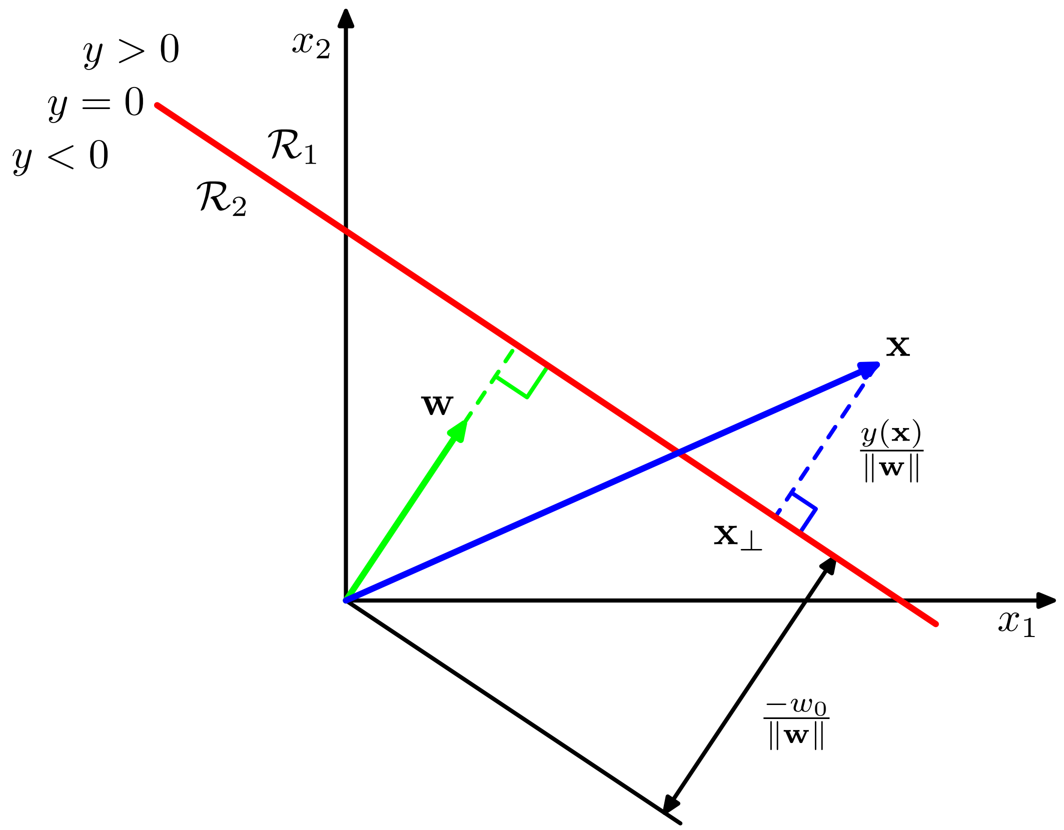


Figure 7.1 from Bishop: The margin is defined as the perpendicular distance between the decision boundary and the closest of the data points, as shown on the left figure. Maximizing the margin leads to a particular choice of decision boundary, as shown on the right. The location of this boundary is determined by a subset of the data points, known as support vectors, which are indicated by the circles.

Review: Geometry of the linear classification

- For a linear predictor function $y(\mathrm{x}) = \mathrm{w}^T\mathrm{x} + b$, recall that:
- w is normal to the decision boundary.
- Distance of x from decision boundary $y = 0$ is proportional to $y(\mathrm{x})$



SVM Basics

- So, the distance of the data points from $y = 0$ is: $\dfrac{t_n y_n}{\|\mathrm{w}\|}$
- Thus, for the closet points, this distance (margin) is: $\dfrac{1}{\|\mathrm{w}\|}$
- To maximise the margin is to minimise $f(\mathrm{w}) = \dfrac{1}{2}\|\mathrm{w}\|^2$
- Subject to constraints that: $1 \leq t_n y_n$

Primal problem

- For the linear separable SVM, the constrained optimisation is specified

$$\min_{\mathbf{w},b} \quad f(\mathbf{w}) = \tfrac{1}{2}\mathbf{w}^\top\mathbf{w} \tag{1}$$

$$\text{subject to} \quad t_i\left(\mathbf{w}^\top\mathbf{x}_i + b\right) \geq 1, \quad i = 1, \ldots, n \tag{2}$$

- Applying the Karush-Kuhn-Tucker conditions, the primal problem is:

$$L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\mathbf{w}^\top\mathbf{w} + \sum_i^n \alpha_i(1 - t_i(\mathbf{w}^\top\mathbf{x}_i + b))$$

$$\alpha_i \geq 0 \qquad \boldsymbol{\alpha} = (\alpha_1 \cdots, \alpha_n)^\top$$

$$1 - t_i(\mathbf{w}^\top\mathbf{x}_i + b) \leq 0$$

$$\alpha_i(1 - t_i(\mathbf{w}^\top\mathbf{x}_i + b)) = 0$$

Slackness conditions

Primal problem

- The primal problem is a convex optimisation problem that can be solved by solvers such as ALGLIB.
- Setting the derivatives of the Lagrangian with respect to the parameters:

$$\begin{aligned} \frac{\partial L}{\partial b} &= 0 \Rightarrow \sum_{i=1}^{n} \alpha_i t_i = 0 \\ \frac{\partial L}{\partial \mathbf{w}} &= 0 \Rightarrow \mathbf{w} = \sum_{i=1}^{n} \alpha_i t_i \mathbf{x}_i \end{aligned}$$

- We solve an example problem to further illustrate.